Project Title
Route Planning Application

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The **Route Planning Application** is a graph-based software system designed to help users find the most efficient route between two or more locations. The application models geographical locations as **vertices (nodes)** and roads or connections between them as **edges**, with distance or travel cost assigned as weights. Based on the given source and destination, the system applies **Data Structures and Algorithms**, particularly graph traversal and shortest-path algorithms such as **Dijkstra's Algorithm**, to determine an optimal route. The application provides the selected path and its total distance, helping users reduce unnecessary travel and choose a suitable route. The project demonstrates the practical implementation of **graphs, priority queues, searching, and shortest-path techniques** in a real-world navigation problem. It also focuses on algorithm efficiency, memory usage, and systematic problem solving. The proposed system can be extended with features such as multiple route suggestions, traffic conditions, real-time location tracking, and estimated travel time, making it more useful for modern transportation and navigation applications.

Signature of the Guide